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Non-Vertical Launch Effects(rev.1)

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Introduction

Much of the analysis done in support of a typical sounding rocket flight is for a perfectly vertical launch. In general this is often a good approximation. But, of course, real flights are never exactly vertical, and, so a means for adjusting vertical flight analysis results to the nearly vertical flight of reality is needed. There are two main problems for which these corrections are needed, dispersions and wind response.

Dispersion

These notes document the derivation of the cell formulas used in Dispersion, an approximate EXCEL based model of sounding rocket trajectory errors. Only the errors in the most important events, apogee altitude and impact point location, are described. A modified version of the classical Impulsive Burn Kepler Jump (IBKJ) model is used with either a Galilean parabola or numerical derivatives, from, say, QROCKET, to map from burnout to subsequent events. "Impulsive burn' is shorthand for the assumption that the powered flight trajectory arc is negligibly small compared to the coasting flight arc. This is sometimes called the "gun shot" approximation.

The event error statistics are assumed to be unbiased. This implies that the predictive models have been suitably validated. Error sensitivities are assumed to be based on linearized (small) perturbations in burnout velocity magnitude (energy errors) and two components of rotation of the burnout velocity vector. Since energy and rotation errors arise from different perturbing effects, they are assumed independent. Furthermore, it can be shown (ref. (2)) that the two rotation errors are also independent. Next, central limit theorem considerations ensure that the event errors have Gaussian distributions. Finally, impact Circular Error Probable (CEP) values at specified capture probabilities are estimated by an approximation due to Reid (ref. (3)).

Finally, this is an extension & correction to the earlier report documented as ref. (1).

Impulsive Burn Kepler Jump Model

The vacuum trajectory parabola results relate burnout velocity **V** and flight path angle $\boldsymbol{\gamma}$ (measured from the horizontal plane up to the velocity vector) to apogee altitude **h*** and impact range **r****:

$$\mathbf{h^* = V^2 \sin^2\gamma \ / 2g}, \text{ and}$$

$$\mathbf{r^{**} = 2\, V^2 \cos\gamma \sin\gamma \,/\, g},$$

where **g** is the acceleration due to gravity.

In Plane Trajectory Errors

Now, consider trajectory errors confined to the plane of the nominal trajectory. If these errors are small compared to their corresponding nominal values, a Taylor series expansion gives a linearized result. For example, the error in impact range is

$$\Delta r^{**} = (\partial r^{**} / \partial V)\,\Delta V + (\partial r^{**} / \partial \gamma)\,\Delta\gamma.$$

Squaring this and averaging over a large ensemble results in

$$var(r^{**}) = (\partial r^{**} / \partial V)^2\, var(V) + (\partial r^{**} / \partial \gamma)^2\, var(\gamma)$$

since **V** and γ are independent. Exactly the same kind of math can be applied to the apogee altitude:

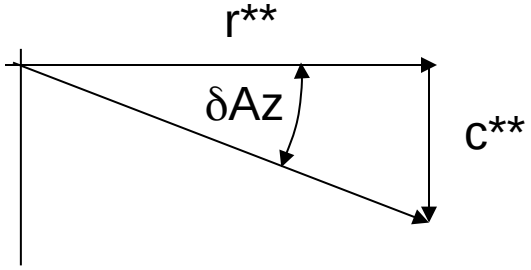
$$var(h^{*}) = (\partial h^{*} / \partial V)^2\, var(V) + (\partial h^{*} / \partial \gamma)^2\, var(\gamma).$$

Note that **V** and γ are the burnout velocity and flight path angle, assumed to occur very near to launch.

Although it is not developed further here, the principal sources of burnout velocity errors are weight errors, specific impulse errors and drag coefficient errors. The principal sources of flight path angle errors are errors in measuring and forecasting the turbulent wind field the rocket flies through, maladjustment of the launcher rail and the misalignment of the thrust vector with respect to the rocket CG. These sources of error are all mutually independent. Thus, the composite errors are just their RSS combination.

Cross Plane Trajectory Errors

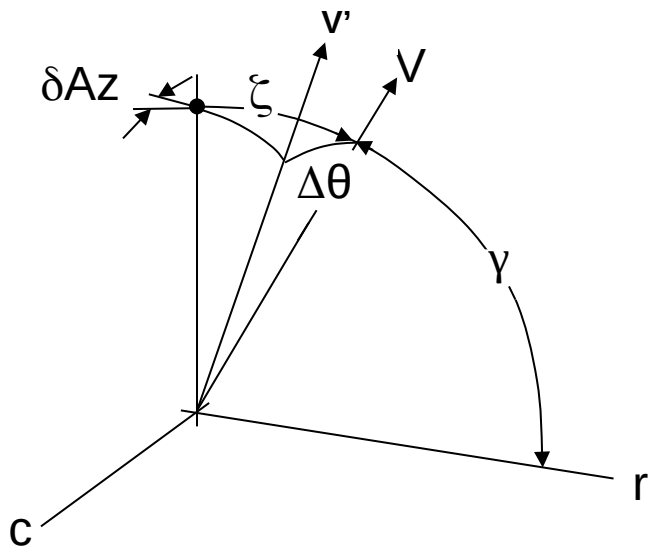
Trajectory errors in a direction normal to the plane of the nominal trajectory are a bit trickier. First, the root errors are the same are for the flight path angle errors above, only these errors are orthogonal to, and independent of, the flight path angle errors. Now, note the trigonometric relation sketched below:



Here, **c**** is the cross plane displacement error of the impact point, nominally at **r****, due to an azimuthal error **δAz.** Thus,

$$c^{**} = r^{**} \tan\delta Az.$$

Next, consider the spherical triangle sketched below in which **V** is the nominal burnout velocity and **V'** is the perturbed velocity:



Now, from spherical trigonometry, we find that

$$\mathbf{\tan\delta Az = \tan\Delta\theta \ / \ \sin\zeta \cong \Delta\theta \ / \ \sin\zeta}$$

Thus,

$$\mathbf{c^{**} = r^{**}\,\Delta\theta \ / \ \sin\zeta}, \text{ or}$$

$$\mathbf{var\,(c^{**}) = (r^{**}/\sin\zeta\ )^2\ var(\Delta\theta)}$$

Next, observe that $\mathbf{\Delta\theta}$ has no influence on apogee altitude errors because $\partial\,\mathbf{h^*} / \partial\,\mathbf{\Delta\theta}$ vanishes as $\mathbf{\Delta\theta \rightarrow 0}$. Lastly, **var(Δθ)** is numerically equal to **var(ζ)** because both are identically driven by unmodeled winds, launcher rail maladjustment and thrust misalignment, thus

$$\mathbf{var\,(c^{**}) = (r^{**}/\sin\zeta)^2\ var(\zeta)}.$$

Dispersions for a Parabolic Trajectory

To put this in perspective, evaluate the sensitivities from the IBKJ model assuming very high boost acceleration (the gun shot approximation):

$$\partial\,\mathbf{r^{**}} / \partial\,\mathbf{V} \approx \mathbf{4\,V\ \cos\zeta\,\sin\zeta\ /\ g},$$

$$\partial\,\mathbf{r^{**}} / \partial\,\zeta \approx \mathbf{2\ V^2\,\cos 2\zeta\ /\ g},$$

$$\mathbf{r^{**}/\sin\zeta \approx 2\,V^2\cos\zeta\ /\ g},$$

$$\partial\,\mathbf{h^*} / \partial\,\mathbf{V} \approx \mathbf{V\sin^2\zeta\ /\ g}, \text{ and}$$
$$\partial\,\mathbf{h^*} / \partial\,\zeta \approx \mathbf{V^2\cos\zeta\ \sin\zeta\ /\ g}.$$

There is evidence that velocity errors increase as the nominal velocity increases…see ref. (2). If the energy errors were redefined in terms of relative velocity error instead of absolute (derivatives taken with respect to $\mathbf{\Delta V / V}$), and $\mathbf{V^2}$ was replaced by

$$\mathbf{V^2 = 2\,g\,h^* / \sin^2\zeta},$$

then

$$\mathbf{V\,\partial\, r^{**} / \partial\, V \approx 8\,h^*\cot\zeta},$$

$$\mathbf{\partial\, r^{**} / \partial\, \zeta \approx 4\,h^*\cos 2\zeta / \sin^2\zeta},$$

$$\mathbf{r^{**} / \sin\zeta \approx 4\,h^*\cos\zeta / \sin^2\zeta},$$

$$\mathbf{V\,\partial\, h^* / \partial\, V \approx 2\,h^*}\text{ , and}$$

$$\mathbf{\partial\, h^* / \partial\, \zeta \approx 2\,h^*\cot\zeta}.$$

This shows clearly that all the above derivatives, and therefore the event dispersions, are proportional to the nominal apogee altitude. Historically, a review of many dispersion analyses lead empirically to this same observation, and that, in turn, lead to the development of ref. (1).

Finally, note that because $\mathbf{\gamma + \zeta = \pi / 2}$, it follows that $\mathbf{d/d\zeta = -\,d/d\gamma}$. The above results, if desired, could be modified to reflect this.

Event Statistics

First, consider the distribution of impact points on the earth's surface. Circular Error Probable is defined as a circle within which a specified fraction of events occur. At one point, the default fraction was implicitly taken to be 50%, but today it's common to say "95% CEP" meaning the circle containing 95% of the events. The CEP concept is directly relevant as a description of impact point errors.

The probability obtained from a surface integral of a bivariate normal distribution over a circular region of radius **R** mathematically defines the CEP. If the in-plane and cross-plane standard deviations were the same, this would be a straightforward application of the circular normal, or Rayleigh, probability distribution. The cumulative Rayleigh distribution also applies to elliptical contours of constant probability density. If the two standard deviations are equal, the Rayleigh result is

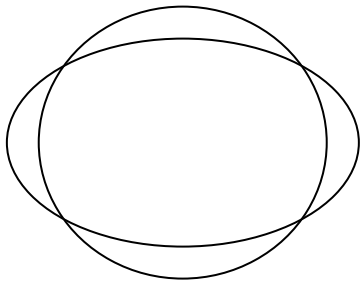
$$\mathbf{Pr(r \le R) = 1 - \exp(-R^2 / 2\sigma^2)}, \text{ where}$$

Pr = Probability the event is captured by the circle,
R = CEP at probability **Pr**, and
$\boldsymbol{\sigma}$ = Standard deviation of one component.

This relation can be solved for **R** to give

$$\mathbf{R = \sigma \sqrt{-2 \log(1 - Pr)}}.$$

But, in general, the two component sigmas are not equal. However, an approximation (ref. (3)) due to R.W.Reid of The Aerospace Corp. provides reasonable answer when the two component standard deviations are roughly the same. Here's the way Reid's "equal area" approximation works:



In the sketch above the ellipse is the true isoprobability density contour, but the circle having the same area will have about the same probability (given by the Rayleigh cumulative distribution). What's happening is that the probability density in the regions above and below the ellipse is less than that on the isoprobability density contour while probability density inside the circle beneath the elliptical "ears" is higher that on the isoprobability density contour. When the ellipse is nearly circular, it can be shown via a Taylor series expansion that the two probability integrals are approximately equal is the circle and ellipse have the same areas. The resulting "one sigma" circle radius $\boldsymbol{\sigma}$ if

$$\boldsymbol{\sigma^2 = \sigma_C \sigma_R}, \text{ where}$$

$\boldsymbol{\sigma_R}$ = In-plane standard deviation, and
$\boldsymbol{\sigma_C}$ = Cross-plane standard deviation.

Putting all the pieces together, the desired CEP formula is

$$\mathbf{R = \sqrt{-2 \log(1 - Pr) * \sigma_C \sigma_R}}.$$

As already mentioned,

$$\boldsymbol{\sigma_R = \sqrt{(V\, \partial\, r^{**} / \partial\, V)^2 var(\Delta V / V) + (\partial\, r^{**} / \partial\, \gamma)^2 var(\gamma)}}, \text{ and}$$

$$\boldsymbol{\sigma_C = r^{**} \sqrt{var(\gamma)} / \sin\gamma}.$$

Finally, the apogee dispersions follow the classical Gaussian distribution with standard deviation $\boldsymbol{\sigma_{h^*}}$**:**

$$\boldsymbol{\sigma_{h^*} = \sqrt{(V\, \partial\, h^* / \partial\, V)^2 var(\Delta V / V) + (\partial\, h^* / \partial\, \gamma)^2 var(\gamma)}}.$$

Unit Wind Effect

The most important source of velocity vector rotation error arises from our inability to measure turbulent gusts, and make the corresponding correction to the launcher angles in a very brief time. For such errors,

$$(\partial\, \mathbf{r^{**}} / \partial\, \zeta)^2 \mathbf{var}(\zeta) = \mathbf{In\ Range\ UWE^2}(\zeta) * \mathbf{var\ (BW)}, \text{ in range, and}$$

$$\mathbf{r^{**2} * var}(\zeta) / \sin^2\zeta = \mathbf{Cross\ Range\ UWE^2}(\zeta) * \mathbf{var\ (BW)}, \text{ cross range, where}$$

BW = Ballistic Wind, and
UWE = Unit Wind Effect.

The wind response varies with QE, ζ, analogously with the rotational dispersion errors. First, the derivative of impact range with respect to flight path angle is

$$\partial\, \mathbf{r^{**}} / \partial\, \zeta = \mathbf{2\, V^2 \cos 2\zeta / g}$$

Then,

$$\mathbf{In\ Range\ UWE}(\zeta) = \mathbf{UWE}(90^o) * \partial\, \mathbf{r^{**}} / \partial\, \zeta\, (\zeta) \,/\, \partial\, \mathbf{r^{**}} / \partial\, \zeta\, (90^o), \text{ or}$$

$$\mathbf{In\ Range\ UWE}(\zeta) = \mathbf{UWE}(90^o) * \cos 2\zeta, \text{ and,}$$

in the same way, $\mathbf{r^{**} / \sin\zeta = 2\, V^2 \cos\zeta / g}$.

Then,

$$\mathbf{Cross\ Range\ UWE}(\zeta) = \mathbf{UWE}(90^o) * \cos\zeta.$$

The nominal apogee altitude & impact range can be used to find an "effective" ζ to be used in the formulae for Unit Wind Effects:

$$\zeta = \tan^{-1}(\mathbf{r^{**}/ h^*}).$$

The data in ref.(4) substantiate this model.

Wind Impact Displacement Algorithm

These results can be used to design a sounding rocket trajectory to have the desired impact location when flown through a known vector wind field. First, decompose the vector winds into components lying in the desired trajectory plane and in a plane normal to the trajectory plane. Next, consider the cross plane wind response. Simulate the rocket response to cross plane winds starting from **QE =** 90°. Adjust the initial **QE** until there is no cross rang impact point displacement. Do the same for the in plane winds, and adjust its **QE** until the desired impact range is achieved. Combine the **QE** adjustments in a vector sense to obtain the launcher **QE** and Azimuth required for a successful launch.

The proof is straightforward. As shown above, the cross plane impact displacement is that for a vertical launch times **cos(ζ) = sin(QE)**. This would be exact if the trajectory were parabolic. Otherwise the **cos(QE)** factor would be something else. But since we are driving this displacement to zero, the multiplicative factor doesn't matter.

References

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3. Reid, R.W., private communication.
4. Schaechter, W., "Sounding Rocket Performance Requirements", Proceedings of the Unguided Rocket Ballistics Conference, ECOM, United States Army Electronics Command, 1967